$$\begin{aligned}\int_0^1 \frac{xe^x}{1+x}\,dx &= \int_0^1 e^x\,dx - \int_0^1 \frac{e^x}{1+x}\,dx \\ &= e - 1 - E\end{aligned}$$

$$\begin{aligned}\int_0^1 \frac{x^2e^x}{1+x}\,dx &= \int_0^1 xe^x\,dx - \int_0^1 e^x\,dx + \int_0^1 \frac{e^x}{1+x}\,dx \\ &= \left[e^x(x-1)\right]_0^1 - (e-1) + E \\ &= E + 2 - e\end{aligned}$$

(i) Let $u = \dfrac{1-x}{1+x},\ x = \dfrac{1-u}{1+u},\ dx = -\dfrac{2}{(1+u)^2}\,du,\ 1+x = \dfrac{2}{1+u}$

$$\begin{aligned}\int_0^1 \frac{e^{\frac{1-x}{1+x}}}{1+x}\,dx &= \int_1^0 e^u \cdot \frac{1+u}{2} \cdot \left(-\frac{2}{(1+u)^2}\right) du \\ &= \int_0^1 \frac{e^u}{1+u}\,du \\ &= E\end{aligned}$$

(ii) Let $u = x^2,\ \dfrac{du}{dx} = 2x,\ \dfrac{dx}{x} = \dfrac{du}{2u}$

$$\int_1^{\sqrt{2}} \frac{e^{x^2}}{x}\,dx = \frac{1}{2}\int_1^2 \frac{e^u}{u}\,du$$

Let $v = x + 1$,

$$E = \int_0^1 \frac{e^x}{1+x}\,dx = \int_1^2 \frac{e^{v-1}}{v}\,dv = \frac{1}{e}\int_1^2 \frac{e^v}{v}\,dv$$

$$\therefore \int_1^{\sqrt{2}} \frac{e^{x^2}}{x}\,dx = \frac{eE}{2}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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